Reading Directions

Not a syllabus. Questions to bring to specific sources.

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May 2026

The programme's questions

The programme reconstructs structure from relations, not spaces. Point-spaces (the real line, Stone spaces, Hilbert spaces) are outcomes of the reconstruction, not its starting material. This is a bet: that the relational description — contexts, reference frames, directed refinement — is the right starting point, and that the structures usually assumed (points, σ-algebras, global states) are *reconstructed* from it.

Three central questions organize the reading:

1. **What must be added to observation to get structure?**

   The persistent question across all phases:

   - Phase 1: observation → dynamics (semigroup)
   - Phase 2–3: observation → geometry (curvature, metric, action)
   - Phase 4–6: observation → probability (CE), realization (descent), value-definiteness (distributivity)

   The reading form: *Where does someone identify the exact condition that forces a valuation to behave — and what structure does that condition live in?*

2. **Reconstruction without points.**

   When can observation be modeled relationally — via contexts and reference frames — rather than by presupposing an underlying set-theoretic structure of points? Numerical identity (point-evaluation) is the degenerate case where all contexts contract to a single point.

   Paper I does this for the Boolean case: probability is reconstructed from directed refinement of finite Boolean algebras, without assuming a sample space. The open question is whether this extends beyond the Boolean case, and if so what plays the role of "directed refinement" when the algebra is not distributive.

   (Sharpened by Zafiris, Ch. 6 of Foundations of Relational Realism: the relational description is not a technical convenience but the starting point; point-spaces are what you reconstruct, not what you assume.)

3. **Local-to-global extension.**

   When does locally defined information (on each Boolean context) determine a global object? What are the *conditions on the site* — not the individual valuations — that control whether local data extend globally?

CE is one instance: local charges always glue finitely additively (Kolmogorov), but gluing to a σ-additive measure requires something extra — and that "something extra" is not a sheaf condition on any known topology (Zafiris 2006, Biesel 2024: dictionary translation, not theorem). Paper II's EA/PR/VDR is another instance: the obstruction to global state extension in the non-Boolean case is non-distributivity rather than failure of σ-additivity.

The general form: both the Boolean and non-Boolean extension problems are local-to-global questions on different categories. What controls the answer is the structure of the site, not the properties of individual states.

(Sharpened by Zafiris, Ch. 6: the gluing conditions on overlapping Boolean reference frames are the structural content; the topology of the site is what does the work.)

1 Caramello — *Theories, Sites, Toposes* (2018)

READ (2026-05-17). Verdict: the bridge technique cannot see CE. σ-additivity is not expressible in finitary geometric logic, so it cannot appear as a quotient in Caramello's framework.

Read Chapters 1, 3–10. The book develops the "bridge technique": given a geometric theory $\mathbb{T}$ of presheaf type with canonical Morita equivalence

$$\mathbf{Sh}(\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{op}, J_{\text{at}}) \;\simeq\; \mathbf{Sh}(\mathcal{C}_{\mathbb{T}'}, J_{\mathbb{T}'}),$$

transfer properties between the syntactic site $(\mathcal{C}_{\mathbb{T}'}, J_{\mathbb{T}'})$ and the semantic site $(\text{f.p.}\mathbb{T}\text{-mod}(\mathbf{Set})^{op}, J_{\text{at}})$ to obtain new results.

The discriminating constraint. A "quotient" of a geometric theory $\mathbb{T}$ means adding a *geometric* sequent — built from finite conjunctions, small disjunctions, existential quantification. σ-additivity requires *countable* universal quantification and countable conjunction: it is not a geometric axiom. So σ-additivity literally cannot be expressed as a quotient of any geometric theory in Caramello's framework. The bridge technique cannot see the finite/countable boundary where CE lives — for the same structural reason that Deligne/Frot cannot: finitary expressiveness.

The infinitary escape. At $\kappa = \omega_1$ (Espíndola's generalization), σ-additivity becomes expressible in κ-geometric logic. But we already established (§8 above) that Property T at ω_1 on the CE site reduces to classical conditions. The bridge either can't see CE (finitary) or reduces it to known content (infinitary).

What the book does provide:

- Theorem 3.2.5 (Duality): bijection between quotients of $\mathbb{T}$ and subtoposes of $\mathbf{Set}[\mathbb{T}]$.
- Chapter 4: coHeyting algebra on subtoposes, Booleanization, DeMorganization.
- Chapter 6 (§6.1.1): canonical Morita equivalences for presheaf-type theories.
- Chapter 10: applications — Fraïssé's theorem topos-theoretically, Galois representations.
- p. 337(b): the theory of decidable Boolean algebras is of presheaf type; f.p. models = finite Boolean algebras; homogeneous models = atomless Boolean algebras. But this is the *algebraic* theory — no measures, charges, or σ-additivity.

No measure-theoretic applications. Caramello has no published work applying the bridge technique to measure theory, valuations, or probability.

Contact with programme: None directly. The bridge technique operates within finitary geometric logic, which cannot express σ-additivity. Sixth instance of the pattern: CE is invisible to every finitary framework tried (Zafiris, Biesel, Frot/Deligne, Espíndola-reduced, Caramello).

Source: Oxford University Press, 2018. ISBN 978-0-19-875891-4.

2 Vickers — *Topology via Logic* (1989)

READ (2026-05-17). Verdict: provides the framework (locales, sublocales, Stone duality) but not the application to measures. The question remains open — requires later literature.

The book is a domain theory / logic textbook. It develops:

- Locales (§5.4): points = frame homomorphisms to **2**; spatialization/localization functors.
- Sublocales (§6.2): = nuclei on frames = frame quotients. The sublocale of "realized points" (principal ultrafilters) is well-defined in this framework.
- Spectral algebraic locales (§§9.2–9.3): coherent presentations, compact coprime opens, Stone duality.
- Stone spaces (§9.5): spectral + Hausdorff ↔ Boolean algebra. Clopens = elements of the Boolean algebra. Booleanization functor. Patch topology.
- ω-algebraicity (§9.4): second countability, ω-chain completions. The finite/countable boundary appears here — finite locales are spectral algebraic (Prop. 9.4.1); second countable ↔ ω-algebraic (Thm. 9.4.6).

What the book does NOT provide:

- No theory of valuations/measures on locales.
- No discussion of σ-additivity or CE in locale-theoretic terms.
- No characterization of "a valuation is supported on the sublocale of principal ultrafilters."

Original question: Is CE expressible as a frame-internal property of a valuation? **Answer: cannot be determined from this book.** The framework is here (locales, sublocales, Stone duality for Boolean algebras), but the theory of valuations on locales is post-1989: Heckmann (1996), Vickers (2008, "A localic theory of lower and upper integrals"). The reading direction remains open but now points to Vickers's later work on localic measure theory, not this textbook.

Useful residue:

- §9.2 Theorem 9.2.2: a frame is *coherent* iff it has a coherent presentation iff $A \cong \mathrm{Idl}(K)$ for a distributive lattice K. This is the frame-level notion behind Deligne/Frot.
- §9.4 Definition 9.4.3: second countability = countable basis. The step from finite to countable basis is exactly the step from coherent to ω-algebraic — parallel to the logical step from finitary to $L_{\omega_1,\omega}$ (Espíndola).
- §6.2 Proposition 6.2.8: sublocales characterized by closure under all meets + Heyting implication condition. The sublocale machinery is available for formulating the support condition — what is missing is the valuation side.

Contact with programme. The framework is now in hand. The open question is whether Vickers's later localic measure theory (2011) provides the valuation-on-locale machinery needed to formulate "μ is supported on the sublocale of principal ultrafilters" as a frame-internal condition. This would be the next reading direction if the CE question is revisited.

Source: Cambridge University Press, 1989. ISBN 0 521 36062 5.

3 Rédei & Summers — "Quantum Probability Theory" (2006)

READ (2026-05-17). Verdict: review paper, Paper II reference.

Reviews quantum probability in the von Neumann algebra framework. Classical probability = abelian von Neumann algebra; quantum = nonabelian. Normal states (= σ-additive) characterized by continuity. The type I/II/III classification controls which probability-theoretic phenomena appear.

Original question: What is the obstruction to extending states across non-Boolean joins? **Answer:** Noncommutativity itself. The type classification refines this but no single algebraic condition beyond "non-distributive + dim ≥ 3" is isolated.

Useful for Paper II: clean reference for the parallel between classical and quantum probability in the operator algebra framework.

Source: arXiv:quant-ph/0601158.

4 Döring & Isham — "What is a Thing?" (2008)

READ (2026-05-17). Verdict: Paper II reference. Daseinisation is descent-adjacent but not CE-relevant.

The topos is $\mathbf{Sets}^{\mathcal{V}(\mathcal{H})^{op}}$: presheaves on the poset $\mathcal{V}(\mathcal{H})$ of commutative subalgebras of $B(\mathcal{H})$. The spectral presheaf $\underline{\Sigma}$ assigns to each Boolean context V its Gelfand spectrum; it has no global elements (= Kochen–Specker). Daseinisation (§5.2–5.3) maps a projection $\hat{P}$ to a sub-object of $\underline{\Sigma}$ by approximating $\hat{P}$ within each Boolean context.

Original question: Does daseinisation give descent from Stone space to realization? **Answer:** Structurally adjacent but different. Daseinisation goes *up* (approximating non-Boolean by Boolean); descent goes *down* (Stone space to realized points). The failure of global sections IS Kochen–Specker, reformulated topos-theoretically — the non-Boolean analogue of CE failure, but with a different obstruction mechanism (noncommutativity vs. non-σ-additivity).

Contact with programme: Paper II directly. Not CE-relevant.

Source: arXiv:0803.0417.

5 Fremlin Vol. 3, Ch. 39 — measurable algebras

READ (2026-05-17). Verdict: answers the question precisely. Weak (σ, ∞)-distributivity is the algebraic CE condition.

391D Theorem (Kantorovich–Vulikh–Pinsker 1950). A Boolean algebra $\mathfrak{A}$ is measurable iff:

1. $\mathfrak{A}$ is Dedekind σ-complete,
2. $\mathfrak{A}$ is weakly (σ, ∞)-distributive,
3. $\mathfrak{A}$ is chargeable (admits a strictly positive finitely additive functional).

391J Theorem (Kelley). $\mathfrak{A}$ is chargeable iff $\mathfrak{A} = \{0\}$ or $\mathfrak{A} \setminus \{0\}$ is a countable union of sets with non-zero intersection numbers.

Original question: Is there a known algebraic condition forcing every charge to be σ-additive? **Answer: Yes — weak (σ, ∞)-distributivity** (plus σ-completeness). KVP 1950 is the theorem. This is NOT a tautological restatement: weak (σ, ∞)-distributivity is a genuine lattice condition on how countable infima interact with arbitrary suprema.

Contact with programme. This is the algebraic answer to Question 3 (local-to-global extension) for the Boolean case. The condition controlling whether local charges extend to a σ-additive global measure is a property of the *algebra* (the site), not the charges (the sections) — exactly what the programme's framing predicts. Weak (σ, ∞)-distributivity is the algebraic face of CE.

Key citation for Paper I: KVP 1950 (via Fremlin 391D) characterizes the exact algebraic boundary. This is what Howson's "missing completeness theorem" looks like in algebraic language.

Source: Fremlin, *Measure Theory* Vol. 3, Ch. 39, freely available at `https://www1.essex.ac.uk/maths/peopl`

6 Vickers — "A localic theory of lower and upper integrals" (2008)

Question while reading: Can "a valuation is supported on the sublocale of principal ultrafilters" be formulated as a frame-internal condition on a localic valuation? Does σ-additivity of the valuation emerge as a locale-theoretic property (e.g., preservation of directed joins, or a continuity condition on the valuation map)?

Why. Vickers (1989) provides the framework (locales, sublocales, Stone duality) but not the application to measures. This later work develops valuations on locales constructively. The programme's question — is CE a frame-internal property? — requires this machinery. If the support condition "μ(sublocale of realized points) = 1" can be expressed without reference to points, it would be the non-circular characterization of coherence.

Contact with programme: CE characterization directly. This is the actual target that Vickers (1989) pointed toward.

Source: *Math. Logic Quarterly* 54(1), 109–123.

7 Howson — "De Finetti, Countable Additivity, Consistency and Coherence" (2008)

READ (2026-05-17). Verdict: identifies the same gap (finite additivity vs. σ-additivity) but does not formalize it. TERMINOLOGICAL WARNING: Howson uses "consistency" and "coherence" as synonyms for de Finetti's criterion (finite additivity / no Dutch book). σ-additivity sits above both. This is opposite to the programme's usage, where consistency = finite additivity and coherence = the stronger condition including σ-additivity.

Howson clarifies de Finetti's position on countable additivity and shows it is internally consistent. Key findings:

The mistranslation. De Finetti's Italian *coerenza* = "consistency" (*consistenza*). His 1972 book uniformly uses "consistency." The English translators (Kyburg) imposed "coherence" against de Finetti's own usage (p. 3). The programme's terminology aligns with de Finetti's original, not the mistranslation.

Consistency = finite additivity, precisely. De Finetti's criterion: P is consistent iff no finite combination of bets at the P-rates guarantees a loss. Characterizes exactly the finitely additive probability functions. Two properties parallel first-order logic: *absoluteness* (consistency depends only on $P[E]$, not on which algebra E sits in) and *compactness* (inconsistency detected by a finite subset). Both fail for σ-additivity (pp. 8–9).

De Finetti's argument against the Dutch Book for σ-additivity (§§3–4). The infinite fair lottery ($p_n = 0$ for all n) is Dutch-Bookable only if postulate (A) — "a finite sum of fair bets is fair" — is extended to countable sums. De Finetti: this extension *entails* σ-additivity, making the Dutch Book argument circular.

Cox $\rightarrow$ finite additivity only (§6, p. 19). Cox's three scale-free axioms + propositional logic $\rightarrow$ finitely additive probability. "Cox's result does not extend to endorsing countable additivity. There are infinitary propositional languages… $L_{\omega_1,\omega}$… but there is no natural extension of Cox's proof which would exploit that additional structure."

The open problem (p. 17). "Something is nevertheless missing, and that is a type of completeness theorem telling us that the rules of probability extend to finite but not countable additivity." This is precisely the gap that Frot and Espíndola formalize: Deligne (finitary completeness) = consistency; CE requires ω_1-completeness.

Contact with programme. Howson (2008) is the primary prior-art citation for the companion note. He identifies the gap (finite additivity has compactness, σ-additivity doesn't) and states the open problem (p. 17: "missing completeness theorem"). The companion note makes this precise via ultraproduct proof. The Frot/Espíndola boundary is the formal version of Howson's philosophical observation: Deligne (finitary completeness) = consistency; CE requires something beyond (ω_1-completeness). Cox's axioms are the right starting material for the programme: they produce finite additivity from structural principles; σ-additivity requires something extra that is not grounded the same way.

Five-traditions unification: DEAD (2026-05-18). Abramsky's contextuality obstruction is compact (finite covers); the σ-additivity gap is non-compact (invisible to finite tests). Category error. Removing Abramsky reduces the thesis to Howson (2008) with fancier coordinates. No new theorem.

Source: *BJPS* 59(1):1–23.

8 Frot — "Gödel's Completeness and Deligne's Theorem" (2013)

READ (2026-05-17). Verdict: confirms the dictionary, places CE.

Proves Deligne's theorem ("coherent topoi have enough points") is equivalent to Gödel's completeness for first-order logic. The dictionary: models = points of the topos (geometric morphisms $\mathcal{E} \to \mathbf{Set}$); consistency = non-triviality; coherent theory = finitary axioms = coherent topos.

Original question: Is CE's failure exactly "the classifying topos lacks enough points"? **Answer: Not exactly.** CE is not expressible in coherent (finitary) logic (confirming Łoś), so Deligne's theorem does not apply. CE lives *above* the coherent level — the guarantee of enough points stops before reaching σ-additivity. The right framing: CE requires a completeness theorem for the *infinitary* fragment where σ-additivity lives.

What connects. Finite additivity IS coherent — Biesel's result (sheaf for finite covers) is the measure-theoretic face of Deligne. σ-additivity requires countable operations, escaping coherence. Frot makes this boundary precise: it is the boundary between finitary and infinitary logic, equivalently between coherent and non-coherent toposes.

Source: arXiv:1309.0389.

9 Espíndola — "Infinitary Generalizations of Deligne's Completeness Theorem" (2017; published *JSL* 85(3), 2020)

READ (2026-05-17). Verdict: places CE at $\kappa = \omega_1$; identifies Property T as the structural condition.

Generalizes Deligne to κ-geometric logic (conjunctions and disjunctions of $< \kappa$ formulas, existential quantification over $< \kappa$ variables). Key results:

- **Theorem 2.11.** If κ is regular with $\kappa^{<\kappa} = \kappa$, κ-geometric theories of cardinality $\leq \kappa$ are complete w.r.t. **Set**-valued models.
- **Corollary 3.1.** Under same hypothesis, every κ-separable topos has enough κ-points.
- **Theorem 3.4.** If κ is weakly compact, every κ-coherent topos has enough κ-points.
- **Corollary 3.3.** κ-coherent theories of cardinality $\leq \kappa$ are complete (= Karp's completeness for $\mathcal{L}_{\kappa,\kappa}$).

Property T (Definition 2.2): an exactness condition on the category — every proper diagram (tree with bar) is completely proper. Combines strong distributivity + dependent choice. At $\kappa = \omega$, property T is automatic (Remark 2.7), recovering classical Deligne. At $\kappa > \omega$ it is a genuine restriction. Rule T (p. 1150) is the syntactic counterpart.

Where CE lives. σ-additivity uses countable universal quantification and countable conjunction $\rightarrow$ expressible in $L_{\omega_1,\omega}$ $\rightarrow$ relevant level is $\kappa = \omega_1$. At this level:

- Theorem 2.11 applies under CH ($\omega_1^{<\omega_1} = \omega_1$).
- Theorem 3.4 requires ω_1 weakly compact (large cardinal axiom).

Reduction check (2026-05-17). Property T at $\kappa = \omega_1$ on the site of finite Boolean subalgebras reduces to σ-completeness of the underlying algebra + uniqueness of the Carathéodory extension. Both classical, both already assumed in any CE setup. No gap for a theorem. Fifth instance of the pattern: CE-adjacent seeds die because the topos/sheaf/logical machinery, applied to the specific site where CE lives, restates classical conditions.

Status. The *coordinates* are genuine (CE lives at $\kappa = \omega_1$ in Espíndola's hierarchy, above Deligne but within Karp). Useful for exposition (Paper I) and for the Caramello reading direction (specifies what her machinery would need to do). Does not produce a seed. The Caramello direction remains open — her Morita equivalence approach is structurally different from "apply Property T to this site."

Source: arXiv:1709.01967.

10 Pták — "Exotic logics" (1987)

READ (2026-05-17). Verdict: populates Square A bottom-right. Obstruction is combinatorial (Greechie pasting), not from the center.

The construction. An *exotic logic* is a σ-orthocomplete OMP that admits finitely additive states but no σ-additive states. Pták constructs these by *Greechie pasting*: start with a collection of Boolean algebras (blocks), identify certain atoms across blocks so that the resulting structure is an OMP, and show that any σ-additive state would require the atoms of one block to sum to more than 1 (by the identifications forcing double-counting).

Original question: Is the obstruction purely from the center, or is there a genuinely non-Boolean mechanism? **Answer:** genuinely non-Boolean. The obstruction is combinatorial: the atom identifications across blocks create contradictory partition-of-unity constraints. Each block individually admits σ-additive states (it is Boolean), but the pasting forces any global state to satisfy incompatible normalization conditions at the σ-additive level. The center of the resulting OMP can be prescribed independently (Theorem 3: any logic embeds into an exotic logic with given center).

Key result (Theorem 1). There exist exotic logics of arbitrary cardinality. The construction is explicit: the pasting diagram and the atom identifications are given.

Contact with programme: Square A directly. Confirms that the OML extension problem (bottom-right cell) has a different character from the Boolean extension problem (top-right, CE): the obstruction is combinatorial/geometric (from the pasting), not analytic (from failure of σ-additivity on a single Boolean algebra). Gleason's theorem is the exception — it requires $\dim \geq 3$ and orthomodularity, and even then only forces σ-additivity on specific algebras.

Source: *Colloquium Math.* 54(1), 1–7.

11 Navara — "Regularity and σ-additivity of states" (1992)

READ (2026-05-17). Verdict: counterexample kills the conjecture that regularity forces σ-additivity. Not an admissibility condition.

The result. Béaver and Cook (1989) proved: on *projection lattices of von Neumann algebras*, every P-regular state (= inner regular w.r.t. finite-dimensional projections) is σ-additive. Navara shows this does NOT generalize: there exist σ-orthocomplete quantum logics (OMPs) on which all states are P-regular yet some are not σ-additive.

Original question: Does regularity force σ-additivity on general quantum logics? Is this the admissibility condition for the right column? **Answer: No.** The Béaver–Cook result is special to von Neumann algebras (where the projection lattice has additional structure: completeness, modularity in finite type, type decomposition). On general σ-orthocomplete OMPs, regularity is strictly weaker than σ-additivity.

The construction. Navara modifies Pták's exotic-logic construction: takes a Greechie pasting that admits only non-σ-additive states, then verifies that all states on this logic are P-regular (because the finite-dimensional sub-logics approximate the full logic sufficiently well for inner regularity, but not for σ-additivity).

Contact with programme: Clarifies the bottom-right cell of Square A. The OML extension problem does *not* have a simple admissibility condition analogous to regularity. The gap between finitely additive states and σ-additive states on OMPs is not bridgeable by regularity alone — the obstruction (Pták's combinatorial pasting) persists even when all states are regular. This confirms that the non-Boolean extension problem is structurally harder than the Boolean one (where weak (σ, ∞)-distributivity does the job via KVP 1950).

Source: *Proc. AMS* 115(2), 427–429.

12 Shultz — "A characterization of state spaces of orthomodular lattices" (1974)

READ (2026-05-17). Verdict: orthomodularity imposes NO geometric constraint on state spaces; but σ-additive state spaces CAN be geometrically degenerate.

Main Theorem (p. 321). If C is a compact convex subset of a locally convex space, then $C \cong S(L)$ (affinely homeomorphic to the state space) of some σ-orthomodular lattice L. If C is a rational polytope, L can be chosen finite.

Construction. Uses Greechie's G-spaces (the same pasting technique as Pták's exotic logics): points + frames with loop constraints. Lemma 1: any G-space yields a σ-OML whose state space is affinely homeomorphic to the weight space $W(X, \mathscr{F})$. Lemmas 2–3: forcing techniques (adjoining points, controlling weight values) to realize arbitrary compact convex sets as weight spaces.

The negative result for σ-states (p. 327). The converse is FALSE for σ-additive states. Take L = Borel subsets of $\mathbb{R}$ modulo Lebesgue-null sets (a σ-Boolean algebra). Then $S_\sigma(L)$ has no extreme points — by Krein–Milman, $S_\sigma(L)$ is NOT affinely equivalent to any compact convex subset of a locally convex space. The σ-state space can be geometrically degenerate in ways the full state space cannot.

Open question (p. 327). How to characterize $S(L)$ if L admits an order-determining set of states? Shultz conjectures there exist compact convex subsets not representable even with this extra assumption.

Contact with programme: Square A, bottom row.

1. Orthomodularity alone imposes NO constraint on state-space geometry — any compact convex set arises. The obstruction to σ-additivity (Pták) cannot be read off from convex geometry.
2. $S_\sigma(L)$ is structurally different from $S(L)$: it can lack extreme points entirely. This is not just "a smaller convex set" — it can fail to be a compact convex set at all in the relevant sense.
3. Greechie pasting is the common tool behind both Shultz (state-space universality) and Pták (exotic logics with no σ-states). The pasting is the fundamental technique for the non-Boolean column.

Source: *J. Combin. Theory A* 17(3), 317–328. DOI: 10.1016/0097-3165(74)90096-X.

13 Epperson & Zafiris — *Foundations of Relational Realism* (2013)

READ (2026-05-17). Verdict: does not address CE.

Read Chapters 6, 8, 9, 10 of the book and Zafiris (2006) *J. Math. Phys.* 47, 092103 (the paper that puts measures on the site). The Grothendieck topology J (epimorphic families) does not grade covers by cardinality — finite, countable, and uncountable covers satisfy J uniformly. The finite/countable boundary where CE lives is invisible. States are finitely additive throughout; σ-completeness is an axiom, not a sheaf condition. No cohomology is computed; the paper proves a *reconstruction* theorem (counit is iso), not an obstruction theorem. The H^1 unification with Abramsky–Brandenburger does not appear.

Original question: Does Zafiris's Boolean localization give a sheaf-theoretic formulation of when local states glue to a global σ-additive state? **Answer: No.**

14 Biesel — "Sheaves of Probability" (2024)

READ (2026-05-17). Verdict: confirms dictionary translation.

Proves measures form a sheaf for finite covers (Thm 8), probability measures likewise (Thm 11). Countable covers explicitly excluded (Remark 7: uniform measures on $\{1, \dots, n\}$ fail to glue to $\mathbb{N}$). Section 5 states explicitly that the results hold for merely finitely additive charges. For countable covers, σ-finiteness is needed — standard measure theory, no hidden structure.

Original question: Is the step from finite to countable covers exactly the CE boundary? **Answer: Yes, but tautologically.** Finitely additive = sheaf for finite covers; σ-additive = sheaf for countable covers. This is a restatement, not a characterization.

Useful residue: Thms 8 and 11 are clean citations for Paper I's CE discussion. Ref [2] (Ross 2012, "All roads lead to violations of countable additivity," *Phil. Studies*) relevant for Howson/de Finetti direction.

Source: arXiv:2401.01968.

Open leads: literature audit results (2026-05-17)

Strategy D

Core question: Does there exist a non-σ-complete, non-atomic Boolean algebra admitting no σ-additive probability?

Key structural clarification. On the *clopen* algebra of a Stone space, σ-additivity is vacuous by compactness: there are no nontrivial countable disjoint families whose join is also clopen.

The real Strategy D question is about *Radon measures* on the Stone space (the Borel/Baire σ-algebra generated by clopens).

Literature findings:

- **Plebanek (2024 survey):** most current secondary source on strictly positive measures on compact spaces. Covers Kelley condition, chain conditions, Fremlin's problem list. Confirms the exact parameter regime (non-basically-disconnected, zero-dimensional, non-atomic) is open.
- **Argyros (1983):** constructs (under CH) a compact, totally disconnected, non-atomic, Radon-measure-free space — but it is basically disconnected (corresponding Boolean algebra is σ-complete). Wrong row of the hierarchy.
- No ZFC construction or obstruction is known for Strategy D's exact conditions. Likely independent.

Verdict: genuinely open, positioned precisely in the landscape. Next step would be consistency constructions (diamond/CH) or reduction to known independence results.

OML extension problem

Core question: What condition on a non-Boolean OML forces every state to be σ-additive (or extendable to a σ-additive measure on the McDonald–Bimbó dual)?

The structural diagnosis. Pták–Pulmannová (1994) prove: if every unital subadditive measure on an OMP is a state, the lattice must be Boolean. This is the killer result — conditions strong enough to force σ-additivity destroy the non-Boolean structure.

Literature findings:

- **Bunce–Wright (1992):** σ-additivity via operator-algebraic structure (JBW-algebras). Positive but narrow: requires completeness close to von Neumann.
- **Chetcuti–Dvurečenskij (2003, 2005):** lattice effects algebras; positive results under strong completeness conditions.
- **Navara (1992):** regularity does NOT force σ-additivity on general OMPs.
- **Recent incremental work:** Voráček–Pták (2023), Burešová–Pták (2023), De Simone–Navara (ongoing). No breakthrough.
- **Wilce (2009), Handbook of Quantum Logic Ch. 24:** confirms Greechie pasting is the universal construction tool for exotic OML state spaces.

Verdict: genuinely open, structurally resistant. The gap between "too weak" and "too strong" appears robust. Needs a new idea (novel pasting technique, or categorical/topos bypass), not more reading.

Context from literature audits (2026-05-15 to 2026-05-17)

The consistency/coherence gap is real and recognized implicitly in five traditions, but never unified under a single name:

- Philosophy of probability (de Finetti/Howson): consistency = finite additivity; σ-additivity is beyond. Howson (2008) confirms and defends de Finetti's position.
- Topos theory (Caramello/Johnstone): "coherent theory" = compactness-controlled; non-coherent = beyond. Frot (2013) proves Deligne = Gödel completeness.
- Infinitary logic (Espíndola 2020): CE lives at $\kappa = \omega_1$ in the κ-coherent hierarchy, above Deligne but within Karp. Property T at ω_1 reduces to classical conditions on the CE site — no gap for a theorem.
- Quantum foundations (Abramsky et al.): contextuality as obstruction to global sections (sheaf cohomology).
- Imprecise probability (Walley): conglomerability = "σ-coherence" in all but name.

- Continuous model theory (Ben Yaacov): sidestep — change the logic so σ-additivity becomes first-order.

No one has unified these. The unification is the opportunity; the risk is that it may be terminological packaging rather than a theorem.

The boundary is now precisely located from three angles:

- **Philosophical:** Howson — Cox's axioms produce finite additivity only; no "natural extension" to $L_{\omega_1,\omega}$ gives σ-additivity.
- **Logical:** Frot — Deligne (enough points for coherent topoi) = finitary completeness = consistency. σ-additivity escapes coherence.
- **Topos-theoretic:** Espíndola — ω_1-coherent completeness exists but Property T on the CE site collapses to classical conditions. The machinery doesn't produce new content at this site.

Earlier findings (2026-05-15 to 2026-05-17):

- The Simpson sublocale conjecture is **dead**: δ_U counterexample kills $\Leftarrow$; compact Hausdorff makes $\Rightarrow$ trivially true.
- The sheaf/gluing approach (CE as descent) confirmed as **dictionary translation** by Zafiris (2006) and Biesel (2024): σ-additivity IS the sheaf condition for countable-partition topology by definition.
- Resolution-dependent charges / fiber defect: **dead**. Collapses to tail mass (Astbury 1981). Interval-charge framework produces no theorem.

Openness as primitive (exploratory, 2026-05-19)

The programme's central question — what forces the transition from finite to countable additivity — discovers incompleteness as a *consequence*: the finitely additive framework is definite and consistent; σ-additivity is the extra structure that fails to be derivable from it. A complementary question: are there frameworks that take openness, indeterminacy, or incompleteness as *primitive*, and if so, do they make contact with the FA/σA boundary?

This section collects reading directions for that question. Each entry carries a specific discriminating question. The question is what determines whether the reference produces a Phase 1 seed or gets parked.

Dzhafarov–Kujala, Contextuality by Default

Primary: E. N. Dzhafarov and J. V. Kujala, "Contextuality-by-Default 2.0," *Lecture Notes in Computer Science* 10106 (2017), 16–32. See also Dzhafarov and Kujala, "Context-content systems of random variables," *Philosophical Transactions of the Royal Society A* 375 (2017), 20160389.

CbD starts from the position that random variables measured in different contexts *lack a joint distribution primitively.* A joint distribution (coupling) is an achievement, not a given. Contextuality is then the failure of a maximally connected coupling to exist. The framework is probabilistic, measure-theoretic, and has a precise notion of "obstruction to global coherence."

Discriminating question: Does CbD's coupling-existence criterion have an analogue for the FA $\to$ σA extension problem? Concretely: can σ-additivity be characterized as the existence of a coupling across refinement contexts — i.e., the existence of a joint distribution on $\bigcup_k \mathcal{G}_k$ compatible with each $\mu|_{\mathcal{G}_k}$? If yes, this would give a non-trivial structural characterization. If the analogy is only verbal ("obstruction to coherence" in both cases but different mathematical content), park it.

Ellerman, Partition Logic and Information — PARKED

Primary: D. Ellerman, "An Introduction to Partition Logic," *Logic Journal of the IGPL* 22 (2014), 94–125. See also Ellerman, "The Logic of Partitions," *Review of Symbolic Logic* 3 (2010), 287–350; and *New Foundations for Information Theory*, Springer (2021).

Status: PARKED (2026-05-19). Read in full. Ellerman builds a complete dual logic where partitions (equivalence relations) play the role that subsets play in Boolean logic. Distinctions ("dits") dualize elements; the partition lattice $\Pi(U)$ is non-distributive; each partition π generates a Boolean core $\mathcal{B}_\pi \cong \mathcal{P}(\pi_{ns})$. "Logical entropy" $h(\pi) = \sum p_i(1 - p_i)$ is the normalized dit-counting measure.

Verdict on discriminating question: The dit/bit translation *is* relabeling entropy. Logical entropy $h(\pi) = 1 - \sum p_i^2 = 1 - \mathrm{Coll}_\mu(\mathcal{G})$, i.e., one minus the collision probability already used in the programme's observational-resolution notes. The partition refinement order recovers the cylinder-algebra filtration. The paper stays entirely in the finite setting — no measures, no charges, no limit behavior, no extension problems. No theorem in partition logic translates to new content about the FA/σA boundary.

Possible re-scoping (not pursued): The non-distributivity of $\Pi(U)$ and the Boolean core construction could speak to the OML extension problem (where distributivity fails), but this would be a different question from the one that motivated the entry.

Walley, Imprecise Probability

Primary: P. Walley, *Statistical Reasoning with Imprecise Probabilities*, Chapman & Hall (1991). For a concise treatment, see M. C. M. Troffaes and G. de Cooman, *Lower Previsions*, Wiley (2014).

Walley takes as primitive a *set* of probability measures (a credal set), not a single measure. Sharp (precise) probability is the special case where the credal set is a singleton. The framework has a natural hierarchy: upper/lower previsions $\to$ coherent previsions $\to$ precise previsions $\to$ σ-additive previsions. Each inclusion is strict. The theory of "natural extension" — the smallest coherent prevision dominating a given assessment — is formally analogous to the problem of extending a finitely additive charge.

Discriminating question: Is there a Walley-type framing where the set of σ-additive extensions of a finitely additive measure on a subalgebra is itself the primitive object, and σ-additivity = singleton extension set? If so, does this give a useful topology or duality on the extension sets (connecting to Choquet theory, Phelps simplex structure)? If the Walley framework only redescribes the same extension problem in different language without new theorems, park it.

Abramsky–Brandenburger, Sheaf-Theoretic Formalism

Primary: S. Abramsky and A. Brandenburger, "The Sheaf-Theoretic Structure of Non-Locality and Contextuality," *New Journal of Physics* 13 (2011), 113036.

Re-scoped. The contextuality application was previously killed: the contextuality obstruction is compact (witnessed by finite covers), while the σ-additivity gap is non-compact (invisible to any finite test). These are different phenomena.

What survives is the *formalism*: a presheaf of local sections (compatible local data), a sheaf condition (existence of a global section), and a cohomological obstruction (H^1 or higher) measuring the failure. The question is whether this formal apparatus can be applied to the Yosida-Hewitt decomposition, where the "local sections" are the restrictions of a finitely additive charge to finite subalgebras and the "global section" is a σ-additive extension.

Discriminating question: Can the purely finitely additive part μ_p of the Yosida-Hewitt decomposition be expressed as a cohomological obstruction in the sense of Abramsky-Brandenburger — i.e., as a non-trivial class in H^1 of a presheaf on a suitable category of finite subalgebras? If so, does the cohomological degree correspond to anything meaningful (e.g., "how far" from σ-additivity)? The key test is whether the compact/non-compact mismatch is fatal to the formalism, not just the application: the Yosida-Hewitt obstruction lives at countable limits, and Čech cohomology of a Stone space is trivial for compact coefficients. If the cohomology is always zero, the formalism doesn't apply.

Sambin, Positive Topology

Primary: G. Sambin, "Some points in formal topology," *Theoretical Computer Science* 305 (2003), 347–408. See also Sambin, *Positive Topology: Completing the Foundational Trilogy* (book in preparation, drafts circulating since 2020).

Sambin's positive topology has two primitives: a *cover relation* (when does a collection of opens cover an open?) and a *positivity predicate* (which opens are "inhabited" — non-trivially instantiated). Points are *derived*, not primitive: a point is a filter satisfying both cover and positivity constraints. The framework is constructive and works without classical logic or the axiom of choice.

Discriminating question: Can the FA/σA transition be viewed as a completion in Sambin's sense, where finitely additive charges live on a "basic" cover and σ-additivity is the positivity predicate forcing concentration on derived points? Concretely: for a Boolean algebra B, is the σ-additive part μ_c of a charge characterized by a positivity condition on the Stone space ("every open containing a point of $\mathrm{supp}(\mu_c)$ is positive"), and if so, does this give a constructive proof of the Yosida-Hewitt decomposition? If the framework only redescribes Stone duality in constructive language, park it.

Supporting philosophical references

The following are not expected to produce Phase 1 seeds directly but may sharpen the question of what "starting from openness" means.

- **Gisin (2021):** N. Gisin, "Indeterminism in Physics, Classical Chaos and Bohmian Mechanics," *Erkenntnis* 86 (2021), 1553–1590. Argues that classical determinism is an artifact of completed infinities in the real-number continuum; intuitionistic reals leave room for genuine indeterminacy. *Relevance:* if σ-additivity depends on completed countable operations, does an intuitionistic treatment change the extension boundary?
- **Linnebo–Shapiro (2019):** Ø. Linnebo and S. Shapiro, "Actual and Potential Infinity," *Noûs* 53 (2019), 160–191. Distinguishes potential from actual infinity with formal precision. See also Hamkins's set-theoretic potentialism. *Relevance:* the FA/σA boundary is exactly where potential infinity (finite additivity: every finite test passes) meets actual infinity (σ-additivity: the countable limit exists). Does the potentialist framework produce a formal distinction beyond the known one?

How to use this document

When reading, annotate in the margins or on a separate sheet:

1. What theorem or definition surprised you.
2. What connected to the programme.
3. What contradicted the knowledge map.
4. What suggested a concrete Phase 1 seed note.

When a reading direction produces a concrete claim worth auditing, extract it to `notes/unsorted/` and run Phase 2.